

Compliance Audit Data Automation

What this project is, and what it does - explained simply

Built by Rahul Mistari | Originally built June 2023, extended through 2026

In one sentence

This project is a small Python program that automatically checks financial trades and client contracts for common paperwork problems - the kind of checking a compliance analyst or auditor would otherwise do by hand.

What problem does it solve?

Financial companies that trade things like interest rate swaps, FX forwards, or credit default swaps are legally required to report those trades to regulators, and to keep complete, accurate records of them. Companies also sign and later end contracts (called mandates) with clients, and those need to be closed out correctly.

In practice, mistakes creep in: a trade gets reported a day late, a required field is left blank, a contract is marked "active" long after it should have ended. None of this is usually fraud - it's ordinary paperwork drift. But regulators still expect it to be caught and fixed, and normally someone has to check for it manually, row by row, in a spreadsheet.

This project automates that checking. It runs a set of rules over sample trade and contract data and reports exactly what it found, with every result traceable back to the specific rule and record that produced it.

What it actually checks

Two groups of records are checked, using two "agents" (independent rule-runners):

Trades (the ledger). Was each trade reported to the regulator on time? Are the required details filled in (what was traded, when, where, at what price, and who the counterparty was)? Is the counterparty properly identified? Do the basic bookkeeping numbers add up, with no duplicate entries?

Client contracts (mandates). If a contract was terminated, does the end date come after the start date? Was the client given enough notice before termination? Does the contract's status field (active / terminated) actually match its dates?

Why three regions - Europe, the US, and Asia-Pacific?

Trade-reporting rules aren't just a European thing. The EU has EMIR and MiFID II. The US has its own version split between the CFTC and SEC, under the Dodd-Frank Act. Australia, Hong Kong, and Singapore each have their own regulators (ASIC, HKMA/SFC, and MAS) with their own rules. Each region gives companies a different deadline to report a trade - the US wants to know almost the same day, Europe gives one business day, and Asia-Pacific gives two.

This project models all three regions at once: every sample trade is tagged with a jurisdiction, and the tool applies that region's actual deadline and required fields when checking it. That's what makes the results a genuine EMIR finding, a genuine Dodd-Frank finding, or a genuine ASIC finding, depending on where the trade happened - not one rule pretending to be global.

EMIR and MiFID II, in one line each

EMIR - does the regulator know this trade happened, and did it find out on time? Think of it as a radar system so regulators can see risk building up across the market.

MiFID II - was the trade done properly, and is the paperwork complete? It's the EU's much broader rulebook covering fair trading and full record-keeping.

What the sample results look like

The tool ships with a synthetic sample dataset (31 trades, 12 client contracts) with a handful of problems deliberately planted so the checks have something to catch. Running it produces:

Checks run	Passed	Had a problem	Needs a person now
191	175	16	13

"Needs a person now" is the subset of problems serious enough (high severity) that someone should manually review them, rather than just note and move on.

Where the checks apply

Region	What it's checking against	Deadline to report a trade
Europe	EMIR and MiFID II (ESMA)	Next business day
United States	Dodd-Frank rules (CFTC and SEC)	Same day
Asia-Pacific	ASIC (Australia), HKMA/SFC (Hong Kong), MAS (Singapore)	2 business days
Internal checks	Company bookkeeping and contract governance, not a specific regulation	Specific regulation

How it's built

The whole thing is plain Python (pandas and NumPy for the data handling), with no external services or paid tools required to run it. Two "validator agents" - one for trades, one for contracts - each apply a set of independent rule classes to every record. Every single rule-and-record combination produces a structured result (pass or fail, severity, a plain-language message), and all of those results are written to an audit trail (CSV and JSON) so nothing is lost and every finding can be traced back to exactly what produced it. A rule-level summary report is generated alongside it.

A results dashboard (a single self-contained HTML file) presents the same results visually, in plain language, for a non-technical reader - including a dedicated page explaining what EMIR and MiFID II actually mean, and a page comparing reporting rules across regions. The same results are also exported as a flat CSV and a formatted Excel workbook, ready to be dropped straight into Tableau Public or Power BI Desktop to build a published dashboard.

What's real, and what's made up

This matters, so it's stated plainly rather than buried: every trade, client name, and contract in this project is fabricated for demonstration. None of it is real data, and no real company or client is referenced anywhere.

What is real: the regulatory deadlines, the required data fields, and the regulator names, which reflect actual public regulatory guidance current as of mid-2026 (EMIR, MiFID II, Dodd-Frank/CFTC/SEC, ASIC, HKMA/SFC, MAS). In short: the data is fake, the rules it's checked against are not. This is a proof of concept meant to demonstrate the automation approach, not a certified compliance tool and not legal

advice.

Who this is for

As a portfolio project, it's meant to show two things at once: an understanding of why these regulatory checks exist and what they actually require, and the ability to automate that kind of manual audit work in code - the bridge between accounting/audit knowledge and data engineering. A companion PDF, "Why These Compliance Rules Exist," covers the reasoning and real-world impact in more depth.

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